Lab 2: Object-Oriented Programming

Lab Objective:

This lab will help students deepen their understanding of object-oriented programming (OOP) in C++. The key concepts covered include the explicit keyword, constructors with initialization lists, constant member functions, and separating the interface from the implementation. Each task builds upon the previous ones, leading to a better understanding of class design and best practices in C++.

Submission:

For each of the tasks, submit your implementation along with the output of the provided main() function.

Task 1: Class Design and Constructors

Objective:

- Design a Student class with private member variables.
- Implement a default constructor, a parameterized constructor, and an explicit constructor to prevent implicit conversions.
- Use an initialization list for constructors.
- Write a member function printDetails() to display the student's information.

Instructions:

1. Create a Student class with the following private member variables:
    - int id - the student's ID.
    - std::string name - the student's full name.
    - float gpa - the student's Grade Point Average.
2. Implement the following constructors:
    - A **default constructor** that sets id to 0, name to "Unknown", and gpa to 0.0.
    - A **parameterized constructor** that takes three arguments (id, name, gpa) and initializes the class members.
    - An **explicit constructor** that takes only id and sets name to "Unknown" and gpa to 0.0.
3. Write a member function printDetails() that displays the student's information.
4. **Test the constructors using the following main() function**.

```
int main() {
    Student defaultStudent;
    defaultStudent.printDetails();

    Student student1(101, "Alice Johnson", 3.9);
    student1.printDetails();

    Student student2(102);
    student2.printDetails();

    return 0;
}
```

Task 2: Constant Member Functions

Objective:

- Ensure immutability by making specific member functions const.
- Add getter and setter functions for the Student class.

Instructions:

1. Add getter functions for id, name, and gpa. These functions should be const since they do not modify any member variables.
2. Add setter functions for name and gpa (not marked as const since they modify class members).
3. Ensure the printDetails() function is also marked as const.
4. **Test the constant functions using the following main() function**.

```
int main() {
    Student student1(103, "Bob Smith", 3.5);
    student1.setName("Robert Smith");
    student1.setGpa(3.6);
    student1.printDetails();

    const Student student2(104, "Charlie Brown", 3.8);
    student2.printDetails();

    return 0;
}
```

Task 3: Constructors with Initialization List

Objective:

- Use initialization lists to initialize member variables when constructing objects.
- Add an additional member variable major to the Student class.

Instructions:

1. Add a new private member variable std::string major to store the student's major.
2. Modify the parameterized constructor to use an initialization list for initializing all member variables, including major.
3. Write a new member function printFullDetails() that prints the student's id, name, gpa, and major.
4. **Test the extended constructor and new function using the following main() function**.

```
int main() {
    Student student1(105, "Diana Prince", 4.0, "Physics");
    student1.printFullDetails();

    student1.setName("Diana P.");
    student1.setGpa(3.95);
    student1.printFullDetails();

    return 0;
}
```

Task 4: Separation of Interface and Implementation

Objective:

- Separate class declarations into a header file (.h) and class definitions into a source file (.cpp).

Instructions:

1. Move the class declaration (including member variables and function prototypes) into a header file (Student.h).
2. Move the function definitions (constructors, printDetails(), etc.) into a separate implementation file (Student.cpp).

3. **Test the separation using the following main() function**.

```
#include "Student.h"

int main() {
    Student student1;
    student1.printDetails();

    Student student2(106, "Clark Kent", 3.8, "Journalism");
    student2.printFullDetails();

    return 0;
}
```